

AMAL JYOTHI COLLEGE OF ENGINEERING KANJIRAPPALLY

B. TECH - COMPUTER SCIENCE AND ENGINEERING

**BTCS2023-27-S6: 23CST362-PROGRAMMING IN PYTHON Project
(Batch – I)**

PROJECT REPORT

Title : TEMPERATURE CONVERTER

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1. Abstract

The Temperature Converter is a Python-based graphical application developed using the Tkinter library. The main objective of this project is to convert a temperature value from one unit into the other commonly used units: Celsius, Fahrenheit, and Kelvin. The application provides a simple and user-friendly graphical interface where users can enter a temperature value and select the unit of measurement. The program then automatically calculates and displays the equivalent temperature values in the other units. Additional features such as temperature condition indication, color change based on temperature levels, graphical visualization using Matplotlib, and background image integration using the Pillow library enhance the usability and visual appearance of the system. This project demonstrates the practical implementation of Python concepts including GUI design, conditional statements, mathematical calculations, and basic data visualization.

2. Problem Description

Temperature is measured using different scales such as Celsius, Fahrenheit, and Kelvin. In scientific studies, weather analysis, and everyday applications, it is often necessary to convert temperature values from one scale to another. Performing these conversions manually can be time-consuming and may lead to calculation errors.

The aim of this project is to develop a simple software application that can automatically convert temperature values between different units. The system should allow users to enter a temperature value, select the unit of measurement, and instantly view the converted values in other temperature scales. The application should also

provide a visually interactive interface that improves user experience and helps users understand temperature conditions more clearly.

3. Libraries & Tools Used (with purpose)

Python

Python is the programming language used to implement the entire application. It provides simple syntax and powerful libraries that make GUI development and mathematical calculations easier.

Tkinter

Tkinter is a built-in Python library used to create graphical user interfaces. It is used in this project to design the application window, labels, buttons, input fields, and layout.

ttk (Themed Tkinter Widgets)

The ttk module provides modern styled widgets for Tkinter applications. In this project, it is used to create the Combobox which allows the user to select the temperature unit.

messagebox

The messagebox module is used to display popup alert messages when invalid input is entered or when errors occur.

Matplotlib

Matplotlib is a Python library used for data visualization. In this project, it is used to generate a bar chart that visually compares the temperature values in Celsius, Fahrenheit, and Kelvin.

Pillow (PIL)

Pillow is a Python imaging library used to load and display images. In this project, it is used to add a background image to the graphical interface, making the application more visually appealing.

4. System Design / Workflow

The workflow of the Temperature Converter system is as follows:

1. The user enters a temperature value in the input field.
2. The user selects the unit of the entered temperature (Celsius, Fahrenheit, or Kelvin) from the dropdown menu.
3. The program reads the input value and converts it to Celsius as an intermediate reference unit.
4. Using standard conversion formulas, the program calculates the equivalent values in Fahrenheit and Kelvin.
5. The converted temperature values are displayed on the graphical interface.
6. A temperature condition such as **Freezing, Cold, Pleasant, or Hot** is displayed based on the Celsius value.
7. The **condition box** changes color according to the temperature level.
8. The user can optionally generate a graph that compares the three temperature values.
9. The background image loaded using the Pillow library enhances the visual appearance of the application.

5. Implementation Details

The project is implemented using Python with the Tkinter library for graphical user interface development. The main window of the application contains widgets such as labels, entry fields, buttons, and a dropdown menu for selecting the temperature unit.

The user inputs a temperature value using the entry widget and selects the unit using the Combobox. The program first validates the input to ensure that it is numeric. If the input is invalid, an error message is displayed using the messagebox module.

The program converts the input temperature to Celsius and then calculates the corresponding Fahrenheit and Kelvin values using mathematical formulas. Conditional statements are used to determine the temperature condition and change the color of the condition label accordingly.

The Matplotlib library is used to generate a bar graph that visually compares the temperature values. Additionally, the Pillow library is used to load and display a background image in the Tkinter window, improving the visual appearance of the interface.

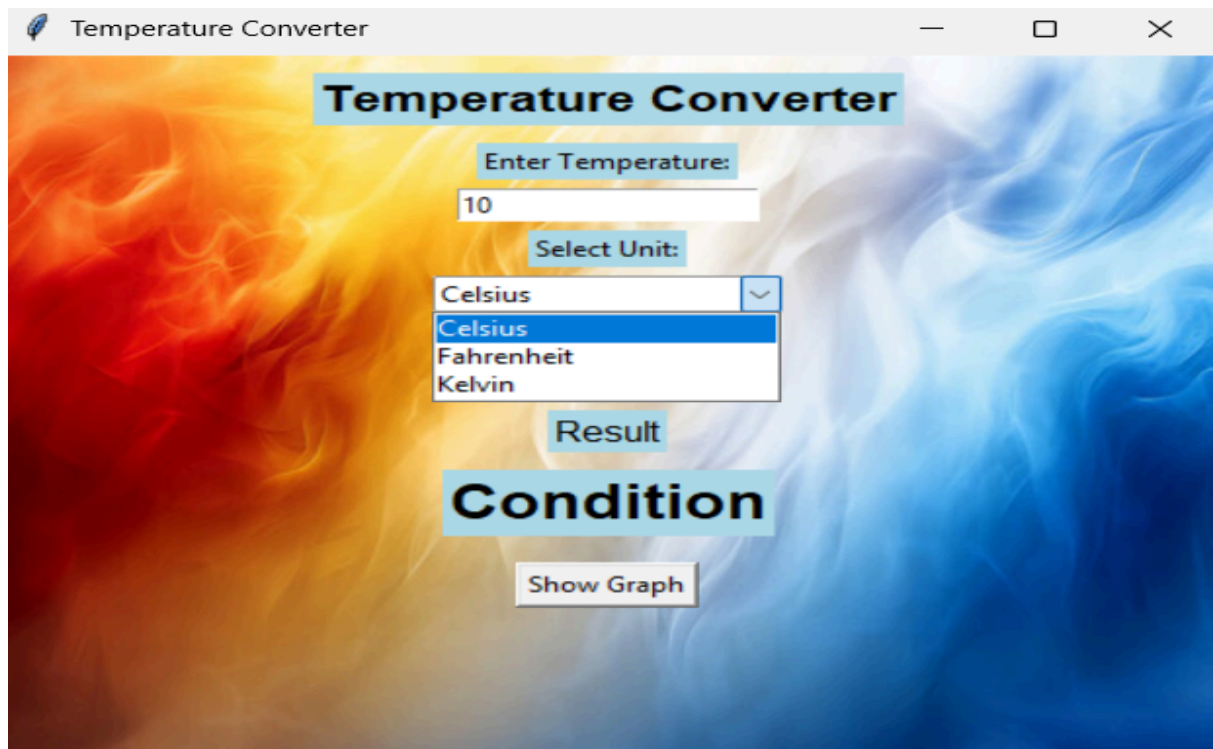
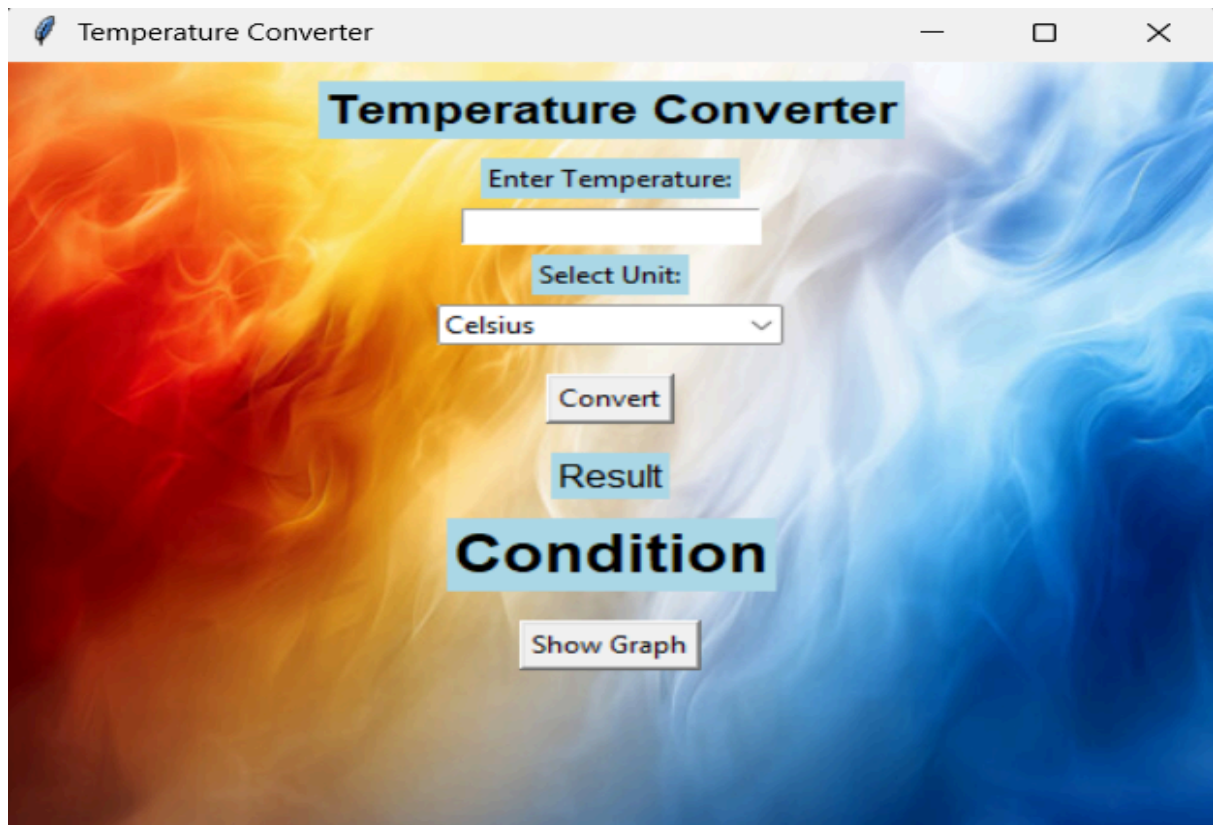
6. Results & Output (with screenshots)

The Temperature Converter application successfully converts temperature values between Celsius, Fahrenheit, and Kelvin. The graphical interface allows users to easily input temperature values and view the converted results instantly.

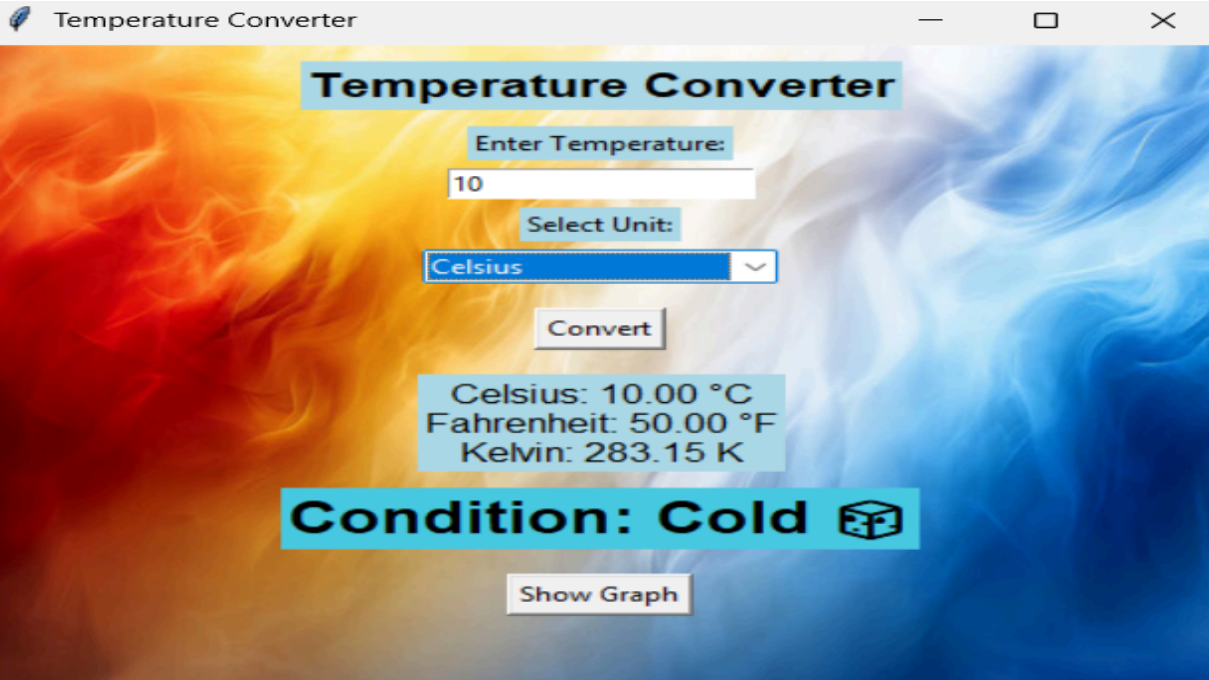
The key outputs of the application include:

- User interface for entering temperature values
- Selection of temperature unit using a dropdown menu
- Display of converted temperature values
- Temperature condition indicator (Freezing, Cold, Pleasant, Hot)
- Color change in the condition box based on temperature
- Background image displayed using the Pillow library
- Graphical representation of temperature values using Matplotlib

1. Main graphical interface of the program



2. Temperature conversion result display



The image shows a window titled "Temperature Converter" with a background of colorful smoke. The interface includes a title bar, a main title, an input field for temperature, a unit selection dropdown, a convert button, a display area for converted values, a condition label, and a show graph button.

Temperature Converter

Enter Temperature:
10

Select Unit:
Celsius

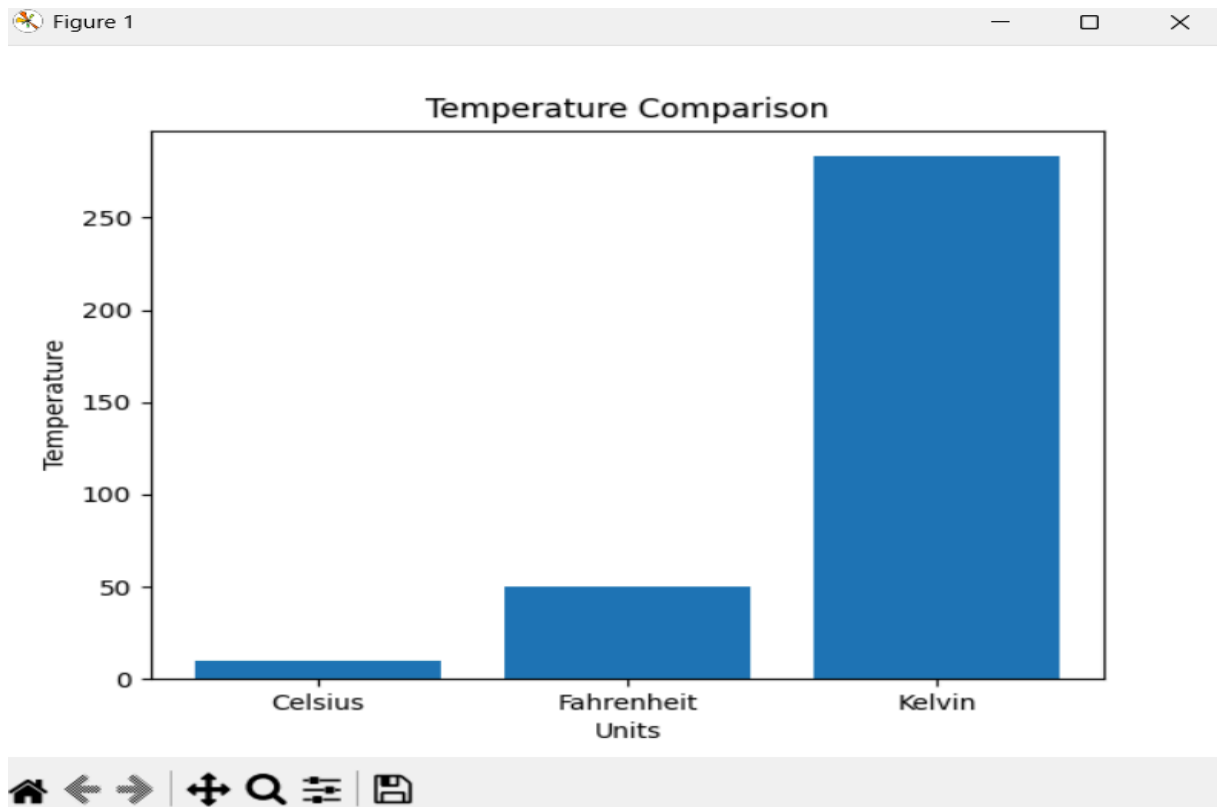
Convert

Celsius: 10.00 °C
Fahrenheit: 50.00 °F
Kelvin: 283.15 K

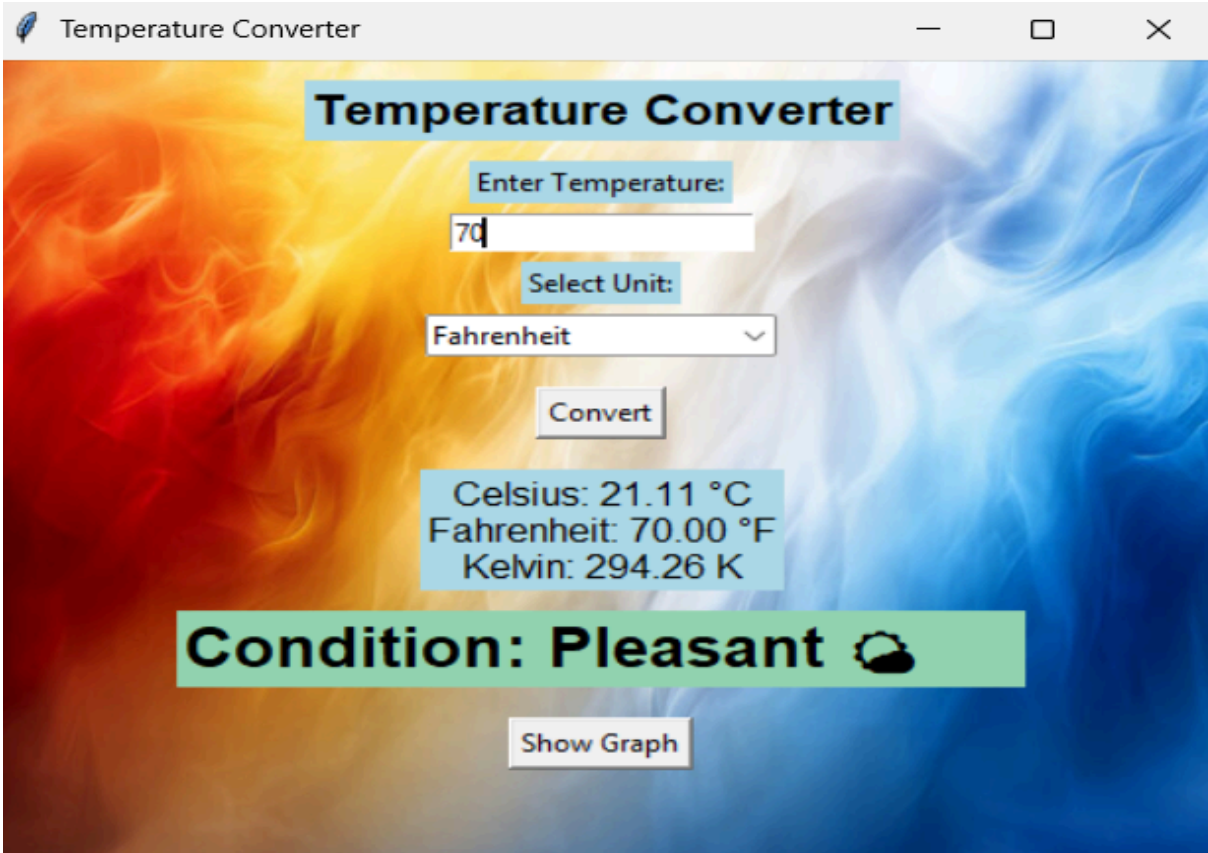
Condition: Cold

Show Graph

3. Graph output generated using Matplotlib



4. Condition box with color change



The image shows a web-based application window titled "Temperature Converter". The background is a colorful, abstract pattern of orange, yellow, and blue. The application features a central form with the following elements:

- Enter Temperature:** A text input field containing the value "70".
- Select Unit:** A dropdown menu currently set to "Fahrenheit".
- Convert:** A button to perform the conversion.
- Results:** A box displaying the converted values:
 - Celsius: 21.11 °C
 - Fahrenheit: 70.00 °F
 - Kelvin: 294.26 K
- Condition: Pleasant** with a sun icon.
- Show Graph:** A button at the bottom.

8. Conclusion & Scope for Mini Project Extension

The Temperature Converter project demonstrates how Python can be used to develop interactive applications with graphical user interfaces. The application successfully performs temperature conversions between Celsius, Fahrenheit, and Kelvin while providing a visually appealing and easy-to-use interface.

The project also helps in understanding key programming concepts such as GUI development, mathematical calculations, conditional logic, and data visualization.

Future improvements for this project may include:

- Adding additional temperature scales such as Rankine
- Implementing real-time conversion while the user types
- Adding a conversion history feature
- Improving the user interface with advanced GUI libraries
- Developing a web-based or mobile version of the temperature converter application